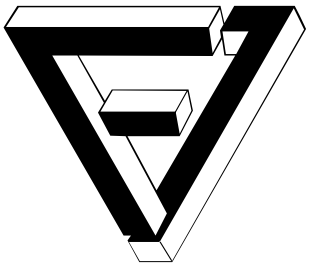
Masaryk University
Faculty of Informatics



Side-Effect Analysis Using a Static Single Use Program Representation

Master's Thesis

Samuel Malec

Brno, 2026

Declaration

Hereby I declare that this thesis is my original authorial work, which I have worked out on my own. All sources, references, and literature used or excerpted during elaboration of this work are properly cited and listed in complete reference to the due source.

During the preparation of this thesis, I used Claude Code (Anthropic, https://claude.ai) to write and review source code, to run and summarise experiments, and to help structure the text. After using this tool, I checked the content and accept full responsibility for it.

Supervisor: Petr Ročkai

Acknowledgements

Write your acknowledgements here.

Abstract

Write your abstract here.

Keywords

side-effect analysis, static single use, linear types, points-to analysis, ahead-of-time compilation, memory optimization, thread escape analysis, lock elision, GraalVM Native Image

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6 Evaluation

This chapter measures what Graal gains from the facts of our analysis when it compiles a program with GraalVM Native Image, and what the facts cost. It counts the loads Graal leaves, times the programs, follows the control flow, and measures the build (Sections 6.2 and 6.3); then it asks why the gain is as small as it is (Section 6.4) and whether the images are correct (Section 6.5). The facts are about memory only: no model of concurrency is used, so every lock and `volatile` access is assumed to order threads. The commands that regenerate the tables are in Appendix D.

6.1 Method

An *image* is a native image built from one program by one builder: Native Image from our fork of Graal, extended with the feature of Chapter 5. The feature takes the facts of Native Image's points-to analysis out of the build, our tool turns them into kill sets and partitions, and Graal's own phases use them. A *kill set* is the set of locations a call may write; without one, Graal assumes that the call may write anything and reloads every field after it. A *configuration* says which facts reach Graal (Table 6.1), and each is compared with the baseline that differs from it in exactly the facts under test.

Native Image runs its points-to analysis in two ways. The *default analysis* names an object by its type; the *allocation-site analysis*, selected with the builder's option `AnalysisContextSensitivity=allocsens`, names it by the place that allocated it. Partitions by object need the finer names; kill sets do not, so we measure them on both. The allocation-site analysis also changes the code Graal produces with none of our facts: on Are We Fast Yet [1] (AWFY) at `-O2` the calls left fall from 1,407 to 1,110 and the loads rise from 943 to 954. We therefore compare images only on the same analysis, and report the price of the allocation-site analysis separately (Section 6.3).

The four small programs were written to contain calls that write little, so the answer is known: they show that the mechanism works, not what a typical program gains. AWFY is the realistic program.

Table 6.1: The configurations. `graal` loads the feature but hands nothing to Graal, so apart from counting it is Native Image as released. `ceiling` leaves out the kill of everything the facts cannot name: it is unsound, an upper bound and not a result. `fields` is also built on allocation sites, against `graal` on allocation sites.

Configuration	Analysis	Facts given to Graal	Compared with
`graal`	default	none	(baseline)
`fields`	default	kill sets, partitions by field	`graal`
`ceiling`	default	kill sets, unsound upper bound	`graal`, `fields`
`graal`	allocation sites	none	(baseline)
`objects`	allocation sites	kill sets and object-field partitions	`graal` on allocation sites

Table 6.2: The programs.

Program	What it is	Used for
`Examples`	34 methods: a call that writes little between a store and a load; 12 of them are controls that must keep their loads	loads
`Signatures`	17 methods, one per rule for what a call's signature says it may touch	loads
`Aliasing`	13 methods: two references that may or may not meet, with 4 controls	loads
`Allocation`	4 methods: a call that allocates enough to collect, which tests the virtual machine state a call kills	loads
AWFY	14 benchmarks over a small collection library of their own, 459 methods, pinned at commit `74306fe`; a `main` constructs each benchmark and prints whether it verified	loads, run time, cost
Native Image's tests	its own JUnit suite: 23 images with 289 tests	robustness

6.2 What Graal Optimizes with the Facts

6.2.1 Loads Left

On the four small programs, kill sets remove 12.5–32.5 % of the loads, and 62 % (29 to 11) in the methods that lose loads at all; the controls keep theirs, as they must. On AWFY they remove 40 of 943 loads (4.2 %) at `-O2` and 54 of 2,565 (2.1 %) at `-O3`, and the same 40 on allocation sites. Partitions by allocation site add nothing on AWFY: `objects` leaves exactly the loads that `fields` leaves, because they split one of its 234 fields. They pay where the program was written for them: on `Aliasing` they remove three loads more (16 to 13).

Table 6.3: Conditions of the measurements.

Item	Setting
Machine	the faculty server pontos04: two AMD EPYC 7371, 64 threads, 1 TB of RAM, Debian 13; the only machine, without hardware counters (`perf_event_paranoid` is 3)
Builder	GraalVM 25.5-dev, built from our fork of Graal on a JDK 25 with JVMCI
Run	2 November 2026, one run of `reproduce.sh` with `THREAD_ESCAPE=off`; tool `df40ed9`, fork `12284dadc70` (control flow: `acb8880`, `8a70f22a6ac`)
AWFY builds	`-O2` as the analysis builds it, and `-O3` with trivial inlining as a user would; they are different compilations, so only the columns of one build compare
Correctness	every image must print exactly what the JVM prints; a build whose output differs is a failure, not a data point
Counts	field loads, stores, and calls left in the program's methods at the end of Graal's low tier; static counts, which loop unrolling multiplies, not run time
Run time	14 benchmarks, 20 iterations, five runs, the first iteration dropped, the median; four cores of one NUMA node, nothing else running; images take turns run by run
Builds timed	`graal` and `fields` four times each, timed together; the first `graal` binary timed twice (A/A) shows the noise
Build time	wall-clock seconds of one build at a time on an empty machine, configurations taking turns, five rounds; median and range, stage times from the builder's log

6.2.2 Run Time

A load that is no longer reloaded saves little by itself; it pays where it enables something more. We therefore time the programs. Table 6.6 gives AWFY at `-O3` on the default analysis.

Queens is the one clear win: −27.5 % to −27.8 % in every build, far outside the noise. Its recursive method `placeQueen` no longer has each call kill four array-reference fields, so their loads leave its loop (Section 6.2.4). **The total** is −0.1 % to −1.4 % (mean −0.8 %), against 0.0 % to +1.1 % for the other builds of `graal`: about 1 %, mostly Queens. Sieve and Storage are faster in all four builds, and we did not investigate why. **Permute** is +7.2 % to +8.5 % slower in every build. Its code changes (Section 6.2.4), but in an earlier experiment that moved all methods to a 16, 64, or 128-byte alignment instead of 32, `fields` was 6.5–7.1 % *faster* than `graal`, so we read it as code placement and not as an effect of the facts. Richards, List, Havlak, and DeltaBlue move by several percent between builds of one configuration, and their ranges for `fields` overlap those of the other builds of `graal`; we make no claim about them.

Table 6.4: Loads left by Graal in the program's methods. The percentage is the change against `graal` on the same analysis. AWFY is built at `-O2` or `-O3`; the two are different compilations, so only the columns of one row compare. The `-O3` build on allocation sites was not run.

	Default analysis		Allocation sites		
Program	`graal`	`fields`	`graal`	`fields`	`objects`
`Examples`	40	27 (−32.5 %)	43	30 (−30.2 %)	30 (−30.2 %)
`Signatures`	16	12 (−25.0 %)	16	12 (−25.0 %)	12 (−25.0 %)
`Aliasing`	16	14 (−12.5 %)	18	16 (−11.1 %)	13 (−27.8 %)
`Allocation`	9	7 (−22.2 %)	9	7 (−22.2 %)	7 (−22.2 %)
Four small programs	81	60 (−25.9 %)	86	65 (−24.4 %)	62 (−27.9 %)
AWFY, `-O2`	943	903 (−4.2 %)	954	914 (−4.2 %)	914 (−4.2 %)
AWFY, `-O3`	2,565	2,511 (−2.1 %)	–	–	–

Table 6.5: The default-analysis loads of the small programs, `graal` to `fields`, by kind of method. *Controls* are the methods named `...Control`: a call in them may write the field, or a reference may alias another, so they must keep their loads. *Unchanged* are the loads of the other methods that do not change (helpers, `main`, inlined monitor code, and in `Signatures` the three methods that cross a monitor, a native call, and a `volatile` access). *Reduced* are the loads of the methods that lose loads. The three add up to the totals of the default analysis in the table above.

Program	Controls	Unchanged	Reduced
`Examples`	16 → 14	10	14 → 3
`Signatures`	–	12	4 → 0
`Aliasing`	5 → 4	9	2 → 1
`Allocation`	–	0	9 → 7
Four small programs	21 → 18	31	29 → 11 (−62.1 %)

6.2.3 The Ceiling

To learn whether more precise facts would help, we built the `ceiling` image, which leaves out the kill of everything the facts cannot name: identity hash codes, class initialization, synchronization, native and unresolved calls. It is unsound by construction, but it prints what the JVM prints on AWFY, and it is the most that precise kill sets could give Graal.

The ceiling removes three times as many loads as `fields` (172 against 54) and is not clearly faster: Queens, the one benchmark that gains, gains the same with `fields`, and Permute and Richards differ by several percent between builds of one configuration anyway. Making the remaining opaque calls precise would not make AWFY faster.

Table 6.6: Run time of AWFY at `-O3`, default analysis, in percent against the first `graal` build (negative is faster). A/A is that binary timed again, the noise. The last two columns are ranges over the three other builds of `graal` and the four builds of `fields`. The six benchmarks not listed are Bounce, CD, Json, Towers, Mandelbrot, and NBody; Mandelbrot and NBody do not change.

Benchmark	`graal`, ms	A/A	other `graal` builds	`fields`, four builds
Queens	23.3	+0.4	−1.8 to −1.5	**−27.8 to −27.5**
Sieve	84.9	−0.8	−5.3 to −4.3	−11.3 to −8.1
Storage	42.3	−0.4	−0.2 to +0.3	−4.8 to −2.7
Permute	27.1	+1.1	−1.1 to +0.4	+7.2 to +8.5
List	39.5	0.0	+1.3 to +2.3	+1.5 to +4.1
Richards	97.6	+0.6	+3.3 to +9.9	+1.7 to +11.5
DeltaBlue	14.4	−0.3	+0.2 to +0.9	−0.1 to +1.7
Havlak	65.7	+0.6	+1.5 to +2.5	−2.1 to +2.4
Other six	27–83	−0.1 to +0.3	−0.4 to +1.3	−0.5 to +1.2
Total	699.6	+0.1	0.0 to +1.1	−1.4 to −0.1

Table 6.7: The ceiling on AWFY at `-O3`, default analysis: the loads left and the run time in percent against `graal`, one build of each; the loads of the four builds timed together differ from these by at most one.

	Loads	In loops	Total	Queens	Permute	Richards
`graal`	2,565	937	–	–	–	–
`fields`	2,511 (−2.1 %)	904	+0.4	−27.9	+6.5	+13.7
`ceiling`	2,393 (−6.7 %)	859	−1.5	−26.7	−7.6	−0.7

6.2.4 Control Flow and Loops

A load that no longer depends on a call may leave a loop, and a branch that depended on it may fold. We count the blocks, control-flow edges, ifs, and loops of every method at the end of Graal's low tier, and for every load whether it is in a loop. The images are built with source positions, so that the same load can be found in two images.

Kill sets barely change the control flow as a whole (6 edges of 29,000), but they change it a lot in three methods. In `Queens.placeQueen` the loop, which contains the recursive call, loses its six loads of the four array-reference fields and then disappears; the method is 2.4 times larger (85 to 207 blocks) and has more edges, because its ifs depend on data. In `Permute.benchmark` the loop disappears too, and most branches with it (49 to 5 ifs); we did not establish why. We read the loops as unrolled by Graal once the loads were gone, but we did not read its decision. Loop-invariant code motion in the strict sense is rare: of the 33 loads that left a loop, 5 appear outside it and 28 are gone.

Table 6.8: Control flow of AWFY at `-O3`, default analysis, summed over the 185 methods the images compile.

	`graal`	`fields`
Blocks	21,197	21,201
Control-flow edges	28,997	29,003
Ifs	8,515	8,514
Loops	602	599
Methods with other control flow	–	9

Table 6.9: The methods whose control flow changes most, `graal` to `fields`. The other five of the nine change by at most 12 edges each.

Method	Blocks	Edges	Ifs	Loops
`Queens.placeQueen`	85 → 207	119 → 299	39 → 97	1 → 0
`Permute.benchmark`	106 → 15	152 → 19	49 → 5	1 → 0
`Permute.permute`	31 → 16	41 → 19	13 → 6	–
`Queens.queens`	120 → 126	169 → 176	–	2 → 1

6.3 Cost

Kill sets add 1.3 seconds (5 %) to the build of AWFY, 1.9 of them in the analysis stage, where the facts are exported and the tool runs. The compile stage does not move (0.3 seconds against a spread of 6.9–8.1), and neither does the memory. The allocation-site analysis costs more than everything we add: `graal` on allocation sites takes 30.4 seconds against 24.1 (+26 %) and its analysis stage 2.7 times as long, and the partitions then add 2.9 seconds (+10 %) and gave nothing on AWFY. The price of the finer analysis grows with the program: javac builds in 67 seconds on the default analysis and in 288 on allocation sites (24 October, three rounds, nothing of ours in either).

6.4 What Blocks the Analysis

The gain on AWFY is small because most of its methods remain opaque: of the 459 methods of the program, 199 still kill everything. Table 6.13 shows what leaving out the kill of each reason would free.

No single reason is enough. Each of the 187 methods that remain after the first three reaches all of an unresolved call, a transition to native code, a raw write, and a snippet, and the reasons form a chain: a method that may build an exception, or call `equals` on an object of unknown type, reaches a lock, then class initialization, then a

Table 6.10: The loads that `fields` removes from AWFY at `-O3`, by where they were. A load that left a loop and appears outside it was hoisted, peeled, or unrolled: the counts cannot tell which. The numbers are changes in the count of loads, `graal` to `fields`.

Loads	Change
in loops	−33
of these, left the loop and appear outside it	5
of these, gone (forwarded, merged, or folded)	28
entered a loop	0
outside loops (the 5 included)	−21
all loads	−54

Table 6.11: Build time of AWFY at `-O3`, median of five rounds in seconds, with the range of the total, and the peak memory. "Added" is the difference of the medians against the configuration above it on the same analysis.

Configuration	Total	Analysis	Compile	Memory, GiB
`graal`	24.1 (23.1–24.8)	4.4	7.4	5.3
`fields`	25.4 (24.8–25.6)	6.3	7.7	5.4
added	+1.3 (+5 %)	+1.9 (+43 %)	+0.3 (+4 %)	+0.1
`graal` on allocation sites	30.4 (30.3–32.6)	11.8	7.4	5.5
`objects` on allocation sites	33.3 (32.6–34.1)	13.7	7.8	5.5
added	+2.9 (+10 %)	+1.9 (+16 %)	+0.4 (+5 %)	0.0

native call. The layers have to go together, and even then AWFY does not run faster (Table 6.7).

6.5 Soundness in Practice

Chapter 4 argues soundness for the model; the proofs are sketches in three places. The evidence of this chapter is of another kind, and Table 6.14 lists it.

This is evidence by controls and examples, not a proof.

6.6 Discussion

Benefit. Graal optimizes a little more with the facts, and in a specific way. Where it must assume that a call writes anything and the call sits in a loop, the kill set lets loads leave the loop and the loop change shape; Queens is that case, with −27.7 %. Elsewhere the gain is a handful of loads (2–4 % of AWFY's) that shows in the total as about 1 %. The small size is not a weakness of the implementation: the ceiling

Table 6.12: Where the added time goes, in seconds (median of five rounds). The export and the tool run inside the analysis stage; applying the kill sets is part of compilation and is summed over the compiler's threads.

Step	`fields`	`objects`
Export of the facts, in the analysis stage	1.7	1.7
Tool: kill sets from the facts, in the analysis stage	0.3	0.3
Applying the kill sets, in compilation	0.1	0.3
Added to the build, total	+1.3	+2.9

Table 6.13: Methods of AWFY (of 459) freed from killing everything when the kill of the named effects is left out of the kill sets (export of 22 October).

Left out	Methods freed
identity hash codes, synchronization, unresolved calls, or other effects, each alone	0
class initialization alone	12
identity hash codes, class initialization, and synchronization together	12
… and unresolved calls	12
… and every other effect	199

removes three times the loads and is no faster, and the opacity of AWFY has several layers, each of which would have to go.

Cost. The facts are cheap on AWFY: kill sets add 5 % to the build. The expensive part is the allocation-site analysis that partitions by object need: it adds 26 % to the build of AWFY and multiplies that of javac by 4.3, and the partitions gained nothing on AWFY. Kill sets on the default analysis are therefore the affordable combination.

Not shown. We did not measure programs larger than AWFY without the concurrency model, so the cost of the facts on them is not reported here, and we did not count loads at run time.

6.7 Threats to Validity

One machine and few programs. Every timing is from one server with an AMD Zen 1 processor and without hardware counters. The realistic program is AWFY; the gains on the other four are by design.

Code placement. Native Image places every method at a multiple of 32 bytes, so a change in the size of one method moves all that follow it. Placement alone moves

Table 6.14: Evidence that the images are correct.

Check	Result
Output	every image of this chapter prints exactly what the JVM prints
Controls that must fail	a method claimed to write nothing gives 171 instead of 177 in `Examples` at `-O3`, so a false claim does not pass unseen
Native Image's own tests	with the feature and `-J-ea -J-esa`, its 23 JUnit images build and all 289 tests pass; the assertions found three defects that the examples did not: disabled flows asked for their objects, calls killed the initialization state, and a JSON escape failed on half a surrogate pair

twelve of the fourteen AWFY benchmarks by 4–13 % with identical instructions, which is above the effect of the facts for all but Queens. We address it by timing four builds of each configuration together and by not citing the benchmarks that vary between builds.

Static counts. A load left is not a load executed, and loop unrolling multiplies counts; `Allocation`'s 66 stores are one store unrolled 64 times. That Graal unrolled the loops of `Queens.placeQueen` and `Permute.benchmark` is our reading of the counts.

One wrong baseline. In the first run of the control-flow measurement one `graal` baseline (AWFY on allocation sites at `-O3`) printed an incorrect result on every run of the binary. Four rebuilds of it and a complete second run, 25 images, were correct. That configuration hands nothing to Graal, so our facts are not the cause; we have not explained it. The tables use the second run.

Single builds and older numbers. The ceiling is one build of each configuration, javac's cost is of 24 October, and the export behind Table 6.13 is of 22 October. The `-O3` build on allocation sites was not run.

7 Related Work

7.1 Side-Effect and Mod/Ref Analyses

7.2 Memory in Compiler Representations

7.3 Escape Analysis and Synchronization Removal

7.4 Analyses for Ahead-of-Time Compilation

8 Conclusion

8.1 Results

8.2 Contribution

8.3 Future Work

Bibliography

[1] Stefan Marr, Benoit Daloze, and Hanspeter Mössenböck. "Cross-Language Compiler Benchmarking: Are We Fast Yet?" In: *Proceedings of the 12th Symposium on Dynamic Languages (DLS)*. 2016.

A Electronic Attachments

B Proofs

C The Dialect of Graal IR in Cthulhu

D Reproducing the Results

This appendix lists what is needed to regenerate the tables of Chapter 6. All commands are in the directories `scripts` and `graal-native` of the tool's repository; the builder is the Graal fork, which the repository contains as a submodule.

D.1 The Tables from One Commit

The script `reproduce.sh` produces the tables of Sections 6.2 and 6.3 in one run. It refuses to start unless the tool and the fork are exactly a commit (no changed tracked files, no untracked files) and the builder is not older than the fork's sources. With `ALLOW_DIRTY=1` it runs for testing and marks the results `NOT-REPRODUCIBLE`. The chapter measures no model of concurrency, which `THREAD_ESCAPE=off` switches off in the builder. The procedure of the run of 2 November, on a machine where nothing else runs, was:

```
# on the development machine, from clean, committed checkouts of the tool and of the fork:
scripts/server/sync-pontos.sh      # copies both trees to the server and stamps them with
    their commits

# on the server:
export JAVA_HOME=$HOME/.mx/jdks/labsjdk-ce-latest-jvmci-25.4-b23_amd64
export PATH=$JAVA_HOME/bin:$HOME/mx:$HOME/.local/bin:$PATH
cd ~/graal/substratevm && mx build # the builder must be newer than the sources
THREAD_ESCAPE=off ~/explicit-effects/scripts/evaluation/reproduce.sh          # loads, time
    , ceiling, cost
THREAD_ESCAPE=off ~/explicit-effects/scripts/evaluation/reproduce.sh control  # control
    flow and loops
```

In `graal-native`, the results are written to a directory of `out/reproduction/`, named by the UTC time, the commit, and `no-thread-escape`: `provenance.txt` (which code), one file per stage, and `SUMMARY.txt`. Table D.1 lists the stages. The first four took 55 minutes on 2 November.

Table D.1: The stages of `reproduce.sh` without a model of concurrency. The first four run by default; `control` runs with `reproduce.sh control`.

Stage	Section	What it runs
`loads`	6.2.1	the configurations on the four small programs and on AWFY at `-O2`, on both analyses, and the counts
`time`	6.2.2	`graal` and `fields` on AWFY at `-O3`, four builds each, timed together with the A/A control
`ceiling`	6.2.3	`graal`, `fields`, and `ceiling` on AWFY at `-O3`: loads left and run times
`cost`	6.3	every configuration of AWFY, five rounds, with the stage times and the peak memory
`control`	6.2.4	control flow, loops, and the place of every load, on the small programs and on AWFY at `-O3`

Table D.2: Scripts for the results outside the stages.

Result	Script
Loads by kind of method	`scripts/evaluation/load_split.py` on the `counts.txt` of two images; `reproduce.sh loads` runs it
javac, price of allocation sites	`scripts/evaluation/cost-javac.sh`, three rounds
Native Image's JUnit tests	`mx native-unittest com.oracle.svm.test` with the feature and `-J-ea`
Placement of code	`programs/awfy/layouts.py` and `time.py`
Controls	`LIE=<method> ./run.sh Examples`; it must go wrong

D.2 Other Measurements

The results that `reproduce.sh` does not regenerate have their own scripts, listed in Table D.2. The loads of the run of 2 November equal those of two earlier runs of the same stages, of 29 and 31 October, to within one load.